Alfredo Navarrete Montes

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EDUCATION

Benemerita Universidad Autonoma de Puebla (BUAP) Puebla, Mexico
Bachelor of Engineering in Information Technologies. *Expected Graduation: December 2026*
Relevant Coursework (completed at University of New Mexico, USA - revalidated): High Performance Computing, Information Systems Development, Computer-based Information Systems, Business Programming Fundamentals, Design of Large Programs.

EXPERIENCE

Health Technologies Laboratory - BUAP Puebla, Mexico
Research Assistant *January 2026 - Present*

- Participated in MentalRiskES 2026 (IberLEF 2026 shared task) on early detection of mental health risk from Spanish social media text, as part of the BUAP research team.
- Achieved 6th place overall in Task 1 and the best early-detection score (R10, MAE 0.889) with the lowest energy consumption among all participating teams.
- Built and ran the competition pipeline in Python, orchestrating multiple LLMs (GPT-4o-mini, GPT-4.1-nano, GPT-5-nano) through the OpenAI API with mock and dry-run workflows.
- Co-authored the resulting paper, accepted for publication in CEUR-WS (indexed in Scopus and DBLP).

University of New Mexico (UNM) - USA Albuquerque, NM
Research Assistant *August 2025 - December 2025*

- Designed a distributed Python scraping pipeline that collected 150k+ Kickstarter campaign records, enabling large-scale analysis of customer behavior.
- Developed automated scrapers in Python to extract structured and unstructured data from multiple online sources.
- Built data preprocessing workflows to clean and merge datasets for analysis and machine learning experiments.
- Collaborated with graduate researchers at the Anderson School of Management to analyze customer behavior and market trends.

HIGH PERFORMANCE COMPUTING PROJECTS

HPC Benchmarking - Center for Advanced Research Computing (UNM) August 2025 - December 2025

- Benchmarked HPL on the Hopper cluster (32 cores per node, InfiniBand, GCC), tuning matrix size N, block size NB, and process grid P x Q through automated Slurm parameter sweeps.
- Raised throughput from 209 to 1,106 GFLOP/s at a fixed rank count by optimizing process grid shape, and reached 2.95 TFLOP/s on 128 MPI ranks.
- Ran CPU-only HPCG experiments on the Easley cluster, scaling from 1 to 128 cores across 4 nodes and analyzing serial fraction and parallel efficiency for a GPU vs CPU study.
- Conducted strong and weak scaling analysis (Amdahl and Gustafson) and co-authored two technical reports in ACM format.

TECHNICAL PROJECTS

Emotion Classification in Spanish (NLP Project)

- Built a transformer-based text classification model using RoBERTuito embeddings.
- Fine-tuned transformer models for Spanish emotion classification, achieving 95% accuracy and 0.95 F1 score on validation data.

TECHNICAL SKILLS

Programming: Python, C++, TypeScript
Distributed Systems & HPC: Slurm, MPI, HPL, HPCG, Linux
Web Development: Angular, Django, REST APIs, MySQL
Data & Machine Learning: NLP, Transformers, LLM APIs, Data Preprocessing, Embeddings
Tools: Git, GitHub

LANGUAGES

Spanish: Native | **English:** Advanced (TOEFL ITP: 560, Duolingo English Test: 120)